

# MD Gouse

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## EDUCATION

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### MS Ramaiah Institute of Technology

*B.E in Computer Science (Cyber Security) GPA: 8.12 / 10*

*Expected OCT 2027*

## PROJECTS

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### Graph-Based Fraud Detection on Bitcoin Transactions | *Python, PyTorch, PyTorch Geometric, GraphSAGE*

- Built a graph-based fraud detection system on the Elliptic Bitcoin transaction dataset (203K nodes, 234K edges).
- Modelled transactions as a graph and trained a 3-layer GraphSAGE network with mini-batch neighborhood sampling to classify transactions as illicit or licit.
- Addressed severe class imbalance (~10% illicit) using inverse-frequency weighted cross-entropy loss computed strictly on the training split.
- Achieved 97% test accuracy with strong recall on the minority (illicit) class.
- Pipeline included custom  $k$ -hop subgraph sampling, block-diagonal batch collation, and cosine annealing LR scheduling.

### Denosing Diffusion Probabilistic Model (DDPM) | *PyTorch, Deep Learning, Generative Models, Computer Vision*

- Implemented a Denosing Diffusion Probabilistic Model (DDPM) from scratch in PyTorch and trained it on the Fashion-MNIST dataset.
- Designed and implemented the forward noise scheduling process and reverse denosing network for probabilistic image generation.
- Modified the standard DDPM objective by incorporating perceptual loss to enhance high-frequency feature reconstruction.
- Achieved sharper, less blurry image samples compared to baseline MSE-trained diffusion models.

## COURSEWORK

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**Mathematics:** Multivariable Calculus, Linear Algebra, Optimization Techniques, Probability and Statistics

**Computer Science:** Data Structures and Algorithms, Operating Systems, Computer Networks, Database Management Systems, Object-Oriented Programming

## TECHNICAL SKILLS

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**Programming Languages:** C, C++, Python, Java

**Frameworks & Libraries:** PyTorch, TensorFlow

**Database Management:** MySQL, MongoDB

**Development Tools & Version Control:** Git, GitHub